8 A ball falls vertically from a great height and rebounds from a horizontal floor. The collision with the floor lasts for a very short time.

When determining the change in momentum of the ball during the collision, its weight is often neglected.

Which statement best explains why this is a valid approximation?

A The weight of the ball does not act while the ball is in contact with the floor.

B The weight of the ball is much smaller than its momentum immediately before the collision.

C The weight of the ball is balanced by the contact force from the floor during the collision.

D The impulse due to the weight of the ball during the collision is much smaller than the impulse due to the contact force from the floor.